MolDStruct MC/MD

Hybrid Monte Carlo / molecular dynamics for X-ray induced ionization and Coulomb explosion

Reference manual

Disclaimer: This manual is AI-generated by claude opus 5.5 with human supervision based on notes, the source code, and simulation runs.

Biophysics group, Uppsala University

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Chapter 1

Introduction

MolDStruct MC/MD simulates what happens to a biomolecule, or any small system of light atoms, when it is hit by an intense femtosecond X-ray pulse. The pulse ionizes the atoms, the released electrons partly escape and partly stay trapped, the sample charges up and explodes, and the ions fly apart in directions that still carry information on the original structure. The model follows all of this in one simulation: the electronic state of every atom, the electrons released, and the nuclear motion.

It is part of the MolDStruct toolbox and is built on GROMACS 4.5.4 [1], extended with an ionization module. Everything GROMACS does still works the same way; the ionization model is switched on with `mdrun -ionize` and controlled by `mcmd-*` options in the `.mdp` file.

1.1 The model in one page

Atoms are classical particles that also carry an electronic configuration: the number of electrons in their K, L and M shells. Each configuration has a charge, and the charges enter the force field.

Electronic transitions (photoionization by the pulse, Auger–Meitner decay, fluorescence) change those configurations. They are sampled every MD step by kinetic Monte Carlo from tabulated rates and cross sections (Chapter 3).

Charge transfer moves electrons between nearby atoms when a classical over-the-barrier criterion allows it (Chapter 4).

The force field is modified so that ionized atoms lose their chemical bonds: bonded terms fade out with the charge, bonds break irreversibly, and the Coulomb interactions between bonded neighbours are restored (Chapter 5).

Released electrons either escape or are trapped by the growing positive charge. The trapped ones form a hot electron cloud, described as a continuous density in thermal equilibrium with the ions: a Poisson–Boltzmann problem solved on a three-dimensional grid every 0.2 fs (Chapter 6). The cloud screens the ions, pulls on them, decides which new electrons escape, and evaporates as the sample expands.

Electron–ion collisions in the cloud ionize further (Chapter 7).

Atomic data (energies, rates and cross sections per configuration) are computed at start-up from a screened-hydrogenic shell model, or read from plain-text tables, which can be generated with XATOM (open source) or from CRETIN models (Chapter 8).

With `mcmd-electrons = none` the electron cloud is switched off: every released electron leaves, and the model is the original MolDStruct model [2].

1.2 How this manual is organized

Chapter 2 gets a first run going. Chapters 3–7 describe the physics, each with the options that control it. Chapter 8 specifies the atomic data, Chapter 9 every output file. Chapters 10 and 11 are the reference: every mdp option and every diagnostic environment variable. The appendices cover the numerics of the electron model (Appendix A), parallel runs (Appendix B), validation and limitations (Appendix C), and units, the source files and the version history (Appendix D).

Basic knowledge of GROMACS is assumed; see the GROMACS documentation for anything not specific to the ionization model.

1.3 Conventions

Inside the model energies are in eV, lengths in nm, and times in ps as in GROMACS — except the pulse options and the electron-model intervals, which are in **femtoseconds**. Charges are in units of e. A configuration is written `K L M`, e.g. `2 4 0` for neutral carbon and `1 4 0` for carbon with a K-shell hole. Potentials are *potential energies of an electron*, in eV.

1.4 Citing

If you use MolDStruct MC/MD, please cite the MolDStruct paper (to be added on publication), and GROMACS 4.5 [1]. The built-in atomic data follow Scott and Hansen [8]; if you use atomic data generated with XATOM, cite XATOM [3].

Chapter 2

Getting started

2.1 Installation

The build is that of GROMACS 4.5.4 with CMake. FFTW 3 (single precision) is required; it is also what the electron model uses for its grid convolutions.

```
cd gromacs-4.5.4-MCMD
mkdir build && cd build
cmake .. -DCMAKE_INSTALL_PREFIX=/path/to/install
make -j8 install
```

Use the `grompp` of this build to write run inputs: the `.tpr` format carries the `mcmd-*` options (format 83), which a stock GROMACS cannot read (and an older stock GROMACS tool would silently misread). Set `GMXLIB` to the build's `share/top` if the force fields are not found.

2.2 The example

`example/` holds a complete run: hen egg-white lysozyme (PDB 1AKI, 1960 atoms) in vacuum, exploded by a Gaussian pulse of 10^{11} photons at 1000 eV, 2.94 fs FWHM, focused to 100 nm.

```
cd example
pdb2gmx -f 1aki.pdb -ff charmm27 -water none
grompp  -f exp.mdp -c conf.gro -p topol.top -o explode.tpr
mdrun   -deffnm explode -v -nt 4 -pd -ionize
```

(`run_example.sh` does the same.) The run is 15 000 steps of 2 as, i.e. 30 fs with the pulse peaking at $t = 0$, and takes about 10 minutes on 4 threads. No atomic data files are needed: the module computes them for 1000 eV at start-up (Section 8.1). The model is the default one; the "Model" block of `exp.mdp` lists its main options.

What to expect. `mcmd-seed = -1` draws a new random seed every run, so the numbers differ by about a percent between runs. Roughly:

- every atom ends up ionized; the mean charge is about 3.5 per atom (hydrogen fully stripped, carbon ∼5.4, nitrogen ∼6.2, oxygen ∼7.1, sulfur ∼13.6);
- about 4000 photoionizations, 800 Auger decays, 2100 electron-impact ionizations and 820 charge transfers;
- of the ∼6900 electrons released, ∼1350 have escaped and ∼5600 are still trapped at 30 fs, at a temperature of ∼18 eV; the radius of gyration has grown 3.6-fold, to ∼5.7 nm.

`tools/mcmd_output.py plot` draws an overview of the whole run (Section 9.4).

2.3 The run summary

At start-up `mdrun` prints a summary of what the module will do, and writes the same block to `md.log` (after GROMACS's own parameter listing): the system, the pulse, the run length and seed, the atomic data found, every process and whether it is on, and the output. For the example:

```
================================================================================
 MolDStruct MC/MD
 System        1960 atoms (H 959, C 613, N 193, O 185, S 10), all neutral at the start
 Pulse         1000 eV, Gaussian, FWHM 2.9435 fs, peak at 0 fs, 1e+11 photons into 100 nm
               (1.27e+13 photons/um^2); bandwidth 0.003
 Run           15000 steps x 2 as = 30 fs; seed 3104721958 (chosen at start, mcmd-seed = -1);
     4 ranks
 Atomic data   built in: screened-hydrogenic shells (K, L, M) at 1000 eV, mcmd-atomic-data =
    hydrogenic
               (written to ./simulation_output/Atomic_data)

 Processes
   photoionization, Auger, fluorescence     on (always)
   charge transfer                          on   hydrogen donors only, H-H off, downhill off
   trapped electrons (Poisson-Boltzmann)    on   mcmd-electrons = pbgrid
       grid                                 s_e 0.25 nm, 2.5 cells/s_e, box <= 3 R, <= 384
    cells,
                                            solve every 0.2 fs, Newton tol 0.01
       evaporation                          on
   ion-ion forces                           all pairs
   electron-impact ionization               on   Lotz, built in (mcmd-coll-data = lotz)
   three-body recombination                 off
   collisional excitation                   off
   field ionization                         off

 Output        simulation_output/: transitions.bin (mcmd-detailed-output = 1),
    charges_over_time.bin every 500 steps;
               autostop off
 Settings      all at the validated production values
================================================================================
```

Settings outside the validated model are listed as warnings instead of the last line — an MD step above 2 as, a PB solver coarser than the converged settings, a non-default smoothing, three-body recombination, collisional excitation, H–H charge transfer, all pairs above 10 000 atoms, harmonic bonds that cannot break, a pulse without photons. The run goes on; the warnings say that the result is not covered by the validation of Appendix C.

2.4 Running your own system

1. Prepare the system as for any vacuum GROMACS run: `pbc = no`, no pressure or temperature coupling during the explosion. Equilibrate it beforehand without `-ionize`.
2. Set `morse = yes` so that `grompp` turns the bonds into Morse potentials, which can break (Chapter 5).
3. Use `dt = 0.000002` (2 as) and the pulse options `mcmd-pulse-*` (Chapter 10); note that the pulse times are in fs.
4. The atomic data are computed for the photon energy of the pulse (H to Ar except He); for other data supply `Atomic_data/` and set `mcmd-atomic-data = files` (Chapter 8).
5. Run `mdrun -ionize`, with `-nt N -pd` for more than one thread (Appendix B).

Ion-ion Coulomb interactions are taken care of by `mcmd-pb-long-range = auto`: all pairs up to 10 000 atoms, a 2 nm cutoff plus the long-range field from the grid above (`grompp` sets the cut-offs and says so).

2.5 How long to run: autostop and the schedule

Most of a run's cost is spent after the pulse, while the sample expands and the ion directions slowly stop turning. Two options shorten that part; neither changes anything before the pulse is over.

Stopping when the directions have settled. `mcmd-autostop = 2` checks the ions every `mcmd-autostop-interval` (5 fs). For the heavy ions (hydrogen is left out: the protons leave first) it takes the angle between the velocity direction now and at the last check, and the median s of these turns over the ions. Late in the explosion the turning rate falls like $1/(t - t_{\mathrm{peak}})^2$, so the turn still to come is extrapolated as

$$\theta_{\mathrm{rest}} = s\,\frac{t - \Delta t_c - t_{\mathrm{peak}}}{\Delta t_c}, \tag{2.1}$$

with Δt_c the check interval. The run ends once $\theta_{\mathrm{rest}} <$ `mcmd-autostop-angle` (2°) and at least `mcmd-autostop-min-time` (60 fs) have passed since the pulse peak (earlier, the turning has not reached the $1/t^2$ regime and the extrapolation undershoots). The decision is taken on the master rank and broadcast; `nsteps` remains the upper limit, and the final frame and checkpoint are written as at a normal end. Every check is a line of `autostop_dirs.txt` (Chapter 9); the stop is announced on stderr.

The extrapolation overestimates: on 250 fs runs (1.3k–21.6k atoms, heavy-atom charge 1.7–6.4, 2 and 4 keV) it predicted 1.7–3 times the turn actually left up to 250 fs. Against runs taken on to 1000 fs (2 keV, mean heavy-atom charge 4–5), the heavy ions turned by a median of 1.1–1.6° after the stop — far below the 7–16° by which the direction of one ion differs between two seeds, but systematic. The ion kinetic energy is still 4–7 % below its final value at the stop. A stopped run is the state at the stop, not the asymptote: compare runs stopped the same way. At 2 keV and 2.6×10^{10} photons into 100 nm (10 fs FWHM), the stop came at a median 85 fs after the start for proteins below 2000 atoms, 105–115 fs for 2000–7000 and 190–205 fs above (pulse peak at 10 fs). `mcmd-autostop = 1`, the older energy criterion ($E_{\mathrm{kin}}/E_{\mathrm{tot}} > 0.99$), says nothing about the directions.

A larger step after the pulse. The 2 as step and the 0.2 fs PB interval are needed while the sample is ionized and the cloud is hot. Afterwards the ions fly almost ballistically through a cooling cloud, and `mcmd-schedule` lets both grow. Each entry `t:dt:pb` (all in fs) starts a stage at time `t` with MD step `dt` and PB interval `pb`; before the first entry `dt` and `mcmd-pb-interval` of the `.mdp` apply. The schedule recommended for production ("IS1c") is

```
mcmd-schedule = 40:0.01:0.2  70:0.1:2  110:0.1:5
```

time	MD step	PB solve every	steps	solves
0–40 fs (the `.mdp`)	2 as	0.2 fs	20 000	200
40–70 fs	10 as	0.2 fs	3 000	150
70–110 fs	100 as	2 fs	400	20
110–250 fs	100 as	5 fs	1 400	28

With a 10 fs pulse peaking at 10 fs it makes runs to 200 fs 2.2–3.5 times cheaper on one thread (1.3k–11k atoms) and moves the ions by 0.16–0.9° (median, heavy ions) and the ion kinetic

energy by −0.4 to −2.6 %, growing with size; the ion maps change 20–100 times less than between two seeds (Appendix C). For other pulses, keep the first change about 30 fs after the pulse peak; a change at 25 fs (15 fs after the peak) doubled the direction error at 11k atoms. Runs beyond 300 fs can add `300:0.5:20`, which changed nothing measurable. With `mcmd-autostop = 2` the gain is smaller (1.3–2 times), since autostop already cuts most of the late part.

`grompp` checks the schedule: the times must increase; every stage must start on a whole number of steps of the stage before and of its own steps from `tinit`; every output interval (`nstenergy`, `nstlog`, `nstxout`, `nstvout`, `nstfout`, `nstxtcout`, `mcmd-charge-output-stride`) must be a whole number of every stage's steps; the integrator must be `md`; and, with photons in the pulse, the first change must come after the pulse peak plus 3σ — changing the step while atoms are still being ionized branches the Monte Carlo history. The end time stays `tinit + nsteps×dt`; `grompp` notes how many steps that takes. The mechanics are in Appendix A, Section A.10.

2.6 Reproducible runs and continuation

The Monte Carlo draws come from a random seed. With `mcmd-seed = -1` (the default) it is drawn from the clock and the process id at start-up, so every run is a different realization — which is what an ensemble needs. The seed is printed in the summary; set `mcmd-seed` to it to repeat a run exactly (on the same number of ranks, Appendix B). Give every member of an ensemble its own seed, or leave it at −1.

Runs can be continued from a GROMACS checkpoint with `mdrun -cpi`. At every checkpoint the module writes `simulation_output/mcionize_state.bin` (the previous one kept as `mcionize_state_prev.bin`): the electronic configurations, the electron cloud's state, the random streams, the reference radius the expansion is measured against, and the length of every output file. On restart the state of the checkpoint's step is read back, the output files are cut back to that step and appended to. The state file is required: the reference radius in particular cannot be recovered from an expanded sample. To extend a finished run, raise `nsteps` with `tpbconv -extend` and continue with `-cpi`.

2.7 The regression test

`test/regression.sh` runs the example for 300 steps with a pinned seed in two cases — `none` (no electron cloud) and `full` (the defaults, on a coarse grid) — and checks that the event counts match the stored references exactly and that the charge is conserved. Then, with `mcmd-atomic-data = files` on the tables the full case wrote, it checks that they reproduce the run bit for bit, that a run without `-ionize` writes no ionization output, that the total energies are required exactly when charge transfer or collisions need them, and that the run is unchanged without the statistical weights. It takes under a minute; `--update` regenerates the references after a deliberate change of the physics. The references are only meaningful for the build and machine that wrote them.

Chapter 3

The atomic model

3.1 Electronic states

The electronic state of an atom is its *shell configuration*: three integers `K L M`, the number of electrons in the $n = 1, 2, 3$ shells (capacities 2, 8 and 18). There are no subshells and no terms or levels within a configuration; whatever an atomic-structure code resolves is summed or averaged per configuration when the data are generated (Chapter 8). An atom starts in its element's ground configuration (table 3.1). Its charge in the MD is $q = Z - (K + L + M)$: at the first step the module replaces the force field's partial charges by these integer ionization charges (all zero at the start), and every transition changes the configuration and the charge.

An atom's element is the one whose standard atomic weight is nearest to its mass, within 0.5 (so argon, 39.948, and calcium, 40.078, are told apart):

element	mass	Z	ground	element	mass	Z	ground	element	mass	Z	ground
H	1.008	1	`1 0 0`	F	18.998	9	`2 7 0`	S	32.06	16	`2 8 6`
Li	6.94	3	`2 1 0`	Ne	20.180	10	`2 8 0`	Cl	35.45	17	`2 8 7`
Be	9.0122	4	`2 2 0`	Na	22.990	11	`2 8 1`	Ar	39.948	18	`2 8 8`
B	10.81	5	`2 3 0`	Mg	24.305	12	`2 8 2`	Ca	40.078	20	`2 8 10`*
C	12.011	6	`2 4 0`	Al	26.982	13	`2 8 3`	Fe	55.845	26	`2 8 16`*
N	14.007	7	`2 5 0`	Si	28.085	14	`2 8 4`	Ni	58.693	28	`2 8 18`*
O	15.999	8	`2 6 0`	P	30.974	15	`2 8 5`	I	126.904	53	`2 8 18`**

Table 3.1: Elements the module knows (`mcmd_elements[]` in `atomic_data.c`); H to Ar have built-in data. * The 4s electrons are counted in M; only with `mcmd-atomic-data = files`. ** A placeholder: right for the force field, not for the ionization physics. Deuterium (mass 2) is hydrogen. Helium is left out: an atom of mass ∼4 is far more likely a heavy hydrogen (`pdb2gmx -heavyh`, mass repartitioning) and is refused rather than taken as helium.

A new element needs a row in that table (mass, symbol, ground configuration) and, above argon or for files, its atomic data.

3.2 Transitions

Three processes are tabulated per configuration, each with its final configuration:

Photoionization (type 2) removes one electron from a shell. The table gives the cross section σ (cm^2) at the photon energy; the rate is $\sigma I(t)$ with the pulse flux $I(t)$ below.

Auger–Meitner decay (type 0) fills a hole from an outer shell and ejects a second electron: the atom loses one electron. The table gives the rate (s^{-1}).

Fluorescence (type 1) fills a hole from an outer shell and emits a photon, which leaves; the charge does not change.

Charge transfer (Chapter 4) and, with the electron cloud, collisional ionization (Chapter 7) are added to these.

3.3 The pulse

The pulse is Gaussian in time, with peak time t_0 (`mcmd-pulse-peak-time`), full width at half maximum τ (`mcmd-pulse-fwhm`), N_{ph} photons (`mcmd-pulse-photons`) in a focal spot of diameter d (`mcmd-pulse-focal-diameter`), uniform over the sample:

$$I(t) = \frac{N_{\mathrm{ph}}}{\pi(d/2)^2}\,\frac{1}{\sigma_t\sqrt{2\pi}}\,\exp\Big[-\frac{(t-t_0)^2}{2\sigma_t^2}\Big], \qquad \sigma_t = \frac{\tau}{2\sqrt{2\ln 2}}, \tag{3.1}$$

in photons $\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$. The fluence $N_{\mathrm{ph}}/(\pi d^2/4)$ is what sets the ionization; the example's 10^{11} photons into 100 nm are 1.27×10^{13} photons/µm^2. The photon energy (`mcmd-pulse-photon-energy`) only enters through the atomic data, which are computed for it (built in) or must be generated for it (files), and through the energy of the photoelectrons. `mcmd-pulse-bandwidth` (relative FWHM, default 0.003) gives each photoelectron a Gaussian jitter of that width in its energy, which matters only for the escape test near threshold; the cross sections are those of the nominal energy.

3.4 The Monte Carlo step

Every MD step Δt, every atom is advanced by a first-reaction kinetic Monte Carlo [6]: for each transition available from its current configuration a waiting time $\delta t = -\ln u/k$ is drawn (u uniform, k the rate, with $I(t)$ for photoionization); the earliest fires if it falls within the time left in the step; then the atom's new configuration offers new transitions, and the procedure repeats until the step is used up. Several transitions can therefore happen to one atom in one step, e.g. a K-shell photoionization followed within femtoseconds by an Auger decay. The Monte Carlo runs on the master rank from one random stream (seeded by `mcmd-seed`, Section 2.6), so the sequence of events does not depend on the number of ranks.

3.5 The released electrons

Each photoionization and Auger decay releases an electron, with kinetic energy from the configurations' total binding energies T (Chapter 8):

$$\varepsilon_{\mathrm{photo}} = E_{\mathrm{ph}} - \big(T_{\mathrm{from}} - T_{\mathrm{to}}\big), \qquad \varepsilon_{\mathrm{Auger}} = T_{\mathrm{to}} - T_{\mathrm{from}}. \tag{3.2}$$

For carbon at 1000 eV a K-shell photoelectron has ∼700 eV and a KLL Auger electron ∼260 eV. Without the electron cloud (`mcmd-electrons = none`) every electron leaves. With it, the electron escapes if ε exceeds the depth of the potential well at its atom, and is trapped in the cloud otherwise (Section 6.7).

Chapter 4

Charge transfer

In a charged sample electrons move between atoms: a highly charged ion pulls an electron over the barrier from a nearby, less charged atom. The model does this with a classical over-the-barrier criterion (`mcmd-charge-transfer = 1`, the default).

4.1 The criterion

For a pair of atoms of different charge, the less charged one is the donor (charge Q_D) and the other the acceptor (Q_A). The electron goes over the barrier when the pair is closer than

$$R_{\mathrm{crit}} = \frac{k_{\mathrm{C}}\bigl(Q_D + 1 + 2\sqrt{(Q_D+1)\,Q_A}\bigr)}{E_{\mathrm{donor}}}, \tag{4.1}$$

with E_{donor} the binding energy of the donor's outermost electron (`energy_levels_X.txt`). Atoms are visited in random order, and each takes the nearest partner that qualifies, within a cutoff of 1.5 nm (a cell list keeps the search linear in the number of atoms).

4.2 Which shell the electron enters

Following Boll *et al.* [5], the electron enters the acceptor vacancy whose binding energy is closest to that of the donor orbital it came from. The binding energy in shell s of configuration c is a difference of total energies,

$$\varepsilon(c, s) = T(c + e_s) - T(c), \tag{4.2}$$

defined for every shell, also for autoionizing configurations. This is why `total_energies_X.txt` is required whenever charge transfer is on. Only shells up to the ground-state occupation are filled, and a vacancy below an occupied shell (a K hole under a filled L shell) is skipped: refilling the shell that absorbed the photon would short-circuit the Auger cascade the rates already describe.

4.3 Variants

`mcmd-ct-h-donor-only = 1` (default): only neutral hydrogen donates, the behaviour of the published model. Hydrogen barely photoionizes at these photon energies and is stripped early by its neighbours.

`mcmd-allow-h-ct` (default 0): allow transfer between two hydrogens. This is resonant and moves no net charge; it is off by default and not fully validated.

`mcmd-charge-transfer-downhill` (default 0): require the transfer to leave the electron more tightly bound, counting the Coulomb energy the pair releases, $(Q_A - Q_D - 1)k_C/R$, by more than 10^{-6} eV. On the lysozyme example it changes the number of transfers by an order of magnitude without changing any observable: charge transfer saturates, and the extra transfers only reshuffle charge the energetics have already fixed.

4.4 Scheduling

The pass over all atoms runs every MD step until nothing has transferred for `mcmd-charge-transfer-idle` steps (default 2000); it then runs only every `mcmd-charge-transfer-recheck` steps (default 100), and every step again as soon as a transfer happens. Because transfer is applied per step rather than at a rate, these two settings change how many transfers happen; they are not pure performance options. A negative idle value never goes idle.

Chapter 5

The modified force field

A force field describes a molecule held together by chemical bonds. Once its atoms are ionized, the electrons that formed those bonds are gone, and the bonded terms would hold the exploding fragments together with a strength they no longer have. With `mdrun -ionize` the bonded and 1–4 interactions are therefore modified as the atoms charge up (`src/gmxlib/bondfree.c` and `src/gmxlib/nonbonded/nonbonded.c`; the switch is the global `moldstruct_altered_ff`). Without `-ionize` the force field is that of stock GROMACS.

5.1 Charges

At the first step the module replaces every atom's partial charge by its ionization charge $q = Z - (K + L + M)$, zero for a neutral atom. From then on the Coulomb interactions act only between ions, with integer charges that change as the atoms ionize. The Lennard-Jones interactions are not modified.

5.2 The ionization factor

Every modified bonded term is scaled by the *ionization factor* of the mean charge $\bar{q}$ of the atoms it involves:

$$f(\bar{q}) = \min\big(1,\ \max(1 - \bar{q},\ 0)\big), \tag{5.1}$$

which is 1 for neutral atoms, falls linearly, and is 0 once the atoms carry on average one charge. Energies and forces are scaled together, so the scaled terms remain conservative at fixed charge.

interaction	GROMACS function	modification under `-ionize`
Morse bond	`morse_bonds`	breaks irreversibly at $\beta(r - b_0) > 4$; then removed and replaced by the pair's C_{12}/r^{12} repulsion; the bare 1–2 Coulomb between the two ions is added
harmonic bond	`bonds`	not modified — use `morse = yes`
angle	`angles`	scaled by $f(\bar{q})$ of the three atoms; explicit 1–3 Coulomb added
Urey–Bradley	`urey_bradley`	angle part scaled by $f(\bar{q})$; the 1–3 spring breaks irreversibly at $r_{13} > 2\,r^0_{13}$, scaled by $f(\bar{q})$ before; explicit 1–3 Coulomb added
proper dihedral	`pdihs`	scaled by $f(\bar{q})$ of the four atoms
improper dihedral	`idihs`	scaled by $f(\bar{q})$ of the four atoms
Ryckaert–Bellemans	`rbdihs`	scaled by $f(\bar{q})$ of the four atoms
CMAP	`cmap_dihs`	scaled by a stepped $f(\bar{q})$ of the five atoms: 1, 0.8, 0.6, 0.4, 0.2, 0 at $\bar{q} <$ 0.19, 0.39, 0.59, 0.79, 1, ≥ 1
1–4 pair	`do_listed_vdw_q`	Coulomb scaled from `fudgeQQ` to full strength, LJ unchanged

Table 5.1: The bonded and 1–4 interactions under `-ionize`.

5.3 Bond breaking

Morse bonds. The bonds should be Morse potentials, $V = D\,[1 - e^{-\beta(r-b_0)}]^2$, which `grompp` makes from harmonic bonds with `morse = yes` (dissociation energies from `share/top/edissoc.dat`); the example does this. A Morse bond breaks permanently once $\beta(r - b_0) > 4$. The threshold is in units of the decay length $1/\beta$, so it lies at the same point of the dissociation curve for every bond type: about 7 % of the peak restoring force is left and 96 % of the well depth has been paid. (A multiple of b_0 would be much more aggressive for short bonds: $2b_0$ leaves 43 % of the peak force on a C–H bond, 20 % on C–C.) After breaking, the Morse term is gone. Because GROMACS's exclusion lists are fixed, the pair stays excluded from the non-bonded interactions; the bond term therefore puts back the pair's own repulsive wall C_{12}/r^{12} (the real C_{12} of the two atom types, without the r^{-6} dispersion), so that broken fragments cannot pass through each other.

The 1–2 Coulomb interaction. Bonded pairs are excluded from the non-bonded Coulomb sum, which is right for partial charges in a molecule but wrong for two ions. The Morse term therefore adds the bare Coulomb interaction of the pair, $k_C q_i q_j / r$, broken or not; its energy is reported in the Morse term of the energy file.

Urey–Bradley 1–3 springs (CHARMM) are harmonic and have no decay length, so their threshold is geometric: the spring breaks permanently at twice its equilibrium length, where the angle it braces is already destroyed.

Harmonic bonds are not modified: they neither weaken nor break. Runs with `-ionize` should use `morse = yes`.

The broken-bond registry. Broken pairs are stored by their global atom indices in a hash set, separately for Morse and Urey–Bradley pairs. It is thread-local: under particle decomposition each rank owns a fixed slice of the bonded interactions, so a bond is only ever evaluated on one rank. It is not written to the checkpoint; after a restart a bond still beyond its threshold re-registers at the first step, so only bonds that broke and came back inside the threshold are forgotten.

5.4 Angles and the 1–3 interaction

Angles are scaled by the ionization factor of their three atoms. As with bonds, 1–3 pairs are excluded from the non-bonded Coulomb sum, so the angle (and Urey–Bradley) terms add the 1–3 Coulomb interaction explicitly, as a genuine potential: energy and force, with the signed charge product.

5.5 The 1–4 interactions

The 1–4 Coulomb interaction of a force field is scaled by `fudgeQQ` (e.g. 0.5–1, depending on the force field) because the bonded terms already describe part of it. Once the bonded terms fade, the 1–4 Coulomb is interpolated back to full strength with the mean charge of the pair:

$$\epsilon_{14} = \epsilon \left[1 - (1 - \texttt{fudgeQQ})\, f(\bar{q}_{ij})\right], \tag{5.2}$$

`fudgeQQ` for neutral atoms and 1 for ionized ones. The 1–4 Lennard-Jones term is not changed.

Chapter 6

Trapped electrons: the Poisson–Boltzmann model

Of the electrons the pulse releases, the fast ones escape and the rest are trapped by the growing positive charge. The trapped electrons thermalize quickly and form a hot, dense nanoplasma [13, 14, 15], which

1. *screens* the ions: it neutralizes the core and reshapes the field that drives the explosion;
2. *ionizes* further by electron impact, at a rate set by its local density and temperature (Chapter 7);
3. *loses* electrons and energy as the well becomes shallower and the fastest electrons evaporate.

With `mcmd-electrons = pbgrid` (the default) the trapped electrons are a continuous density $n(\boldsymbol{x})$ on a three-dimensional grid that follows the sample. Every `mcmd-pb-interval` (0.2 fs) the density is recomputed as the equilibrium distribution of N classical electrons at one temperature T, in the field of the ions and of the electrons themselves. T is not a parameter: it follows from the electrons' total energy, which is booked event by event. The density gives the forces on the ions, the depth that decides whether a new electron escapes, the local density for collisions, and the electrons that evaporate. How it is discretized and solved is in Appendix A. With `mcmd-electrons = none` there is no cloud: every released electron leaves.

6.1 Why a continuum, and why Boltzmann

Three lengths decide the description: the sample radius R (a few nm, growing), the Debye length $\lambda_D = \sqrt{\varepsilon_0 T/(e^2 n)}$ (0.05–0.5 nm for $n \sim 10^2$–$10^3\,\mathrm{nm}^{-3}$ and $T \sim$ 10–100 eV), and the interionic distance (0.1–0.2 nm in the intact protein). With $\lambda_D \ll R$ the electrons behave like a conductor: they screen the field inside the sample, leaving a quasi-neutral core with the net positive charge in a thin outer layer [16]. This follows from the Poisson–Boltzmann equation; it does not follow from a prescribed electron profile.

A classical Boltzmann gas at one temperature rests on three assumptions:

1. *fast thermalization*: the electron–electron collision rate, 10^{15}–$10^{16}\,\mathrm{s}^{-1}$ [17], is faster than the explosion;
2. *no degeneracy*: the thermal wavelength $\lambda_{\mathrm{th}} \approx 0.12$ nm at 10 eV gives $n\lambda_{\mathrm{th}}^3 \lesssim 1$ except in the deepest wells;
3. *mean field*: correlations beyond the mean field are neglected; bound states are the atomic Monte Carlo module's business.

The electrons follow the ions on the scale of the inverse plasma frequency, $\omega_p^{-1} \lesssim 0.1$ fs, much faster than the ions move, so the density can be recomputed as an equilibrium rather than

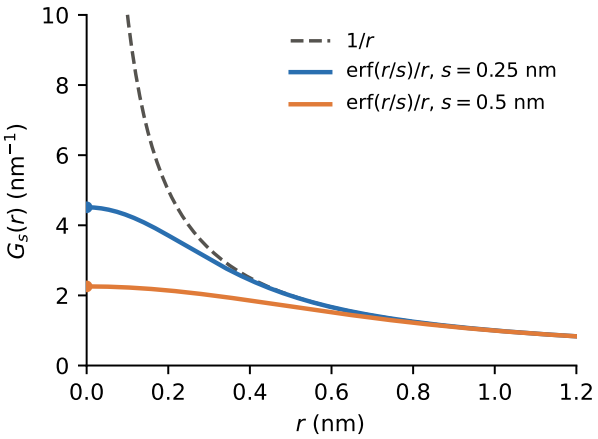


Figure 6.1: The smoothed Coulomb kernel $G_s(r) = \operatorname{erf}(r/s)/r$ against $1/r$; the dots are $G_s(0)$. The model's $s_e = 0.25\,\mathrm{nm}$ is the blue curve.

integrated in time.

6.2 The free energy

In the mean-field approximation the Helmholtz free energy of N classical electrons at temperature T, in the field of fixed ions, is

$$F[n] = -\int n\,\Phi\,\mathrm{d}^3x + \frac{1}{2}\iint n(\boldsymbol{x})\,K(\boldsymbol{x}-\boldsymbol{y})\,n(\boldsymbol{y})\,\mathrm{d}^3x\,\mathrm{d}^3y + T\int n\Big[\ln\frac{n}{n_*}-1\Big]\mathrm{d}^3x, \qquad \int n\,\mathrm{d}^3x = N, \tag{6.1}$$

with the ions' attraction $\Phi(\boldsymbol{x}) = k_\mathrm{C}\sum_a q_a G_s(|\boldsymbol{x}-\boldsymbol{x}_a|)$, the electron–electron kernel $K(\boldsymbol{r}) = k_\mathrm{C}G_s(r)$, and the entropy of an ideal gas (n_* only shifts F by a constant). It is the classical, local limit of Mermin's finite-temperature density-functional theory [18], without exchange and correlation. Minimizing F at fixed N with a chemical potential μ gives

$$n(\boldsymbol{x}) = n_*\,e^{(\mu - V(\boldsymbol{x}))/T}, \qquad V = -\Phi + K * n, \tag{6.2}$$

the Boltzmann factor of the self-consistent potential. Since $\nabla^2(1/r) = -4\pi\delta$, this is the Poisson–Boltzmann equation $\nabla^2 V = 4\pi k_\mathrm{C}(\tilde{\rho}_\mathrm{ion} - \tilde{n})$, $n \propto e^{-V/T}$ (with smoothed sources), solved with open boundaries. Hot, the cloud follows the shallow total well over the whole sample; cold, it neutralizes the interior and leaves the net charge in a sheath of thickness $\sim \lambda_D$ at the surface.

F is strictly convex: the entropy has second variation $T/n > 0$, the Hartree term is positive definite (its kernel's Fourier transform $\hat{K} = (4\pi k_\mathrm{C}/k^2)\,e^{-k^2s^2/4}$ is positive), and the ion term is linear. There is exactly one minimizer, and the solver (Appendix A) converges to it from any positive starting density.

6.3 The smoothed Coulomb kernel

Every charge — ions and electron cells — is a Gaussian, and two Gaussians interact through

$$G_s(r) = \frac{\operatorname{erf}(r/s)}{r}, \qquad G_s(0) = \frac{2}{s\sqrt{\pi}}, \tag{6.3}$$

the splitting that also underlies Ewald summation [19, 20]. G_s equals $1/r$ to 10^{-7} beyond $4s$ and is finite at $r = 0$ (Fig. 6.1).

There are two reasons to smooth. On a grid of spacing Δx a cell is a blob of width $\sim \Delta x$; treating it as a Gaussian with $s = 2\Delta x$ makes the grid sums smooth approximations of the

integrals. And physically, a classical Boltzmann gas collapses into a point ion ($e^{k_C q/rT}$ is not integrable at $r \to 0$): the electron's quantum size, 0.1–0.2 nm at 10–50 eV, blurs the singular well, and the bound part of the well is the atomic module's. The model uses

$$s = \max\bigl(2\Delta x,\ s_e\bigr), \qquad s_e = \texttt{mcmd-pb-smoothing} = 0.25 \text{ nm}, \tag{6.4}$$

everywhere the kernel appears: the ion and electron potentials, the escape depth, distant ions and the long-range ion force. Once the grid resolves s_e ($\Delta x \le s_e/2$), refining it no longer changes the physics. The charge, the escape and the ion directions do not depend on s_e; the ion kinetic energy does (Appendix C).

6.4 The temperature

The temperature is that of an ideal gas, $T = \frac{2}{3}K/N$, with K the electrons' kinetic energy — the usual equipartition definition, since the potential energy is explicit. K is not tracked directly: the model conserves the electrons' *total* energy $E_{\text{tot}} = K + U$, with the electrostatic energy $U(\boldsymbol{n}) = \int n\,[-\Phi + \frac{1}{2}K * n]$ (electrons with ions, and half their mutual repulsion; the ion–ion energy is GROMACS's). Since the density depends on T, T and $\boldsymbol{n}$ are found together, from

$$g(T) = E_{\text{tot}} - U(\boldsymbol{n}_T) - \tfrac{3}{2}NT = 0, \tag{6.5}$$

where $\boldsymbol{n}_T$ minimizes F at T. A hotter cloud spreads into regions of higher potential energy, so g decreases monotonically and has one root, found by a safeguarded secant iteration (Appendix A).

Temperature floor. A cloud can cool below any sensible temperature late in a weak explosion (in 1B7U at 8 keV and 10^7 photons K reaches zero with 6000 electrons still trapped). The solve, the evaporation and three-body recombination therefore use $\max(T, T_{\min})$, with $T_{\min} =$ `mcmd-te-min` (1 eV): the density never collapses into a zero-temperature well, and recombination, which grows as n_e^2/T (Section 7.2), stays finite. The actual $T = \frac{2}{3}K/N$ keeps its own bookkeeping and may go below $T_{\min}$, or below zero when E_{tot} is less than the potential energy of the floored density; the floor adds no energy. Electron-impact ionization uses the actual T; it vanishes at small T by itself ($e^{-\Delta E/T}$). `screening_log.txt` has both temperatures, and `mdrun` reports how long T was below the floor. With `mcmd-te-min = 0` (and in a `.tpr` older than version 81) the release model applies: the solve is floored at 1 eV, evaporation and recombination use the actual T, and when $E_{\text{tot}} - U$ falls below zero, K is set to zero and E_{tot} raised to match (energy created, logged in `pb_log.txt`). The production runs used that model; it made a difference only in the case above (6–28 keV created after 64 fs, $\le$5 % of the ion kinetic energy).

6.5 Energy bookkeeping

Tracking E_{tot} instead of K means that the adiabatic cooling of an expanding cloud, and the heating when an ion's charge grows under it, appear by themselves in $K = E_{\text{tot}} - U$. E_{tot} changes only by explicit exchanges. Write $V_e = K * n$ for the electrons' repulsion from the last solve, scaled by $\varsigma = N/N_{\text{solve}}$ for electrons added or lost since, and

$$D_k = k_C \sum_{b \ne k} q_b\, G_s(|\boldsymbol{x}_k - \boldsymbol{x}_b|) - \varsigma\, V_e(\boldsymbol{x}_k) \tag{6.6}$$

for the depth of the well at atom k: the binding energy of an electron at rest there.

1. **Work on the ions.** The cloud's force on the ions does work W per step, taken from the electrons: $\Delta E_{\text{tot}} = -\sum_a \boldsymbol{F}_a \cdot \boldsymbol{v}_a \Delta t$.
2. **A trapped photo- or Auger electron** of energy ε at atom k: $\Delta E_{\text{tot}} = \varepsilon - D_k$.

3. **Collisional ionization** at k with binding energy ΔE_{b}: a cloud electron pays it, and the freed electron starts at rest at k: $\Delta E_{\mathrm{tot}} = -\Delta E_{\mathrm{b}} - D_k$.
4. **A change of an ion's charge** by Δq changes its interaction with the cloud; the kinetic energy does not change at that instant: $\Delta E_{\mathrm{tot}} = -\Delta q\, \varsigma V_e(\boldsymbol{x}_k)$.
5. **Evaporation** of δN electrons with kinetic energy δE from mean depth $\bar{D}$: $\Delta E_{\mathrm{tot}} = -(\delta E - \delta N \bar{D})$.
6. **A grid rebuild** changes U for numerical reasons only; the change is moved into E_{tot} so that K does not jump.

An electron that escapes at birth never joins the cloud; it only raises the net charge. `energy_ledger.txt` and `pb_log.txt` record these terms (Chapter 9).

6.6 Forces on the ions

An ion of charge q_a has potential energy $-q_a \varsigma V_e(\boldsymbol{x}_a)$ in the cloud, so the cloud pulls on it with

$$\boldsymbol{F}_a = q_a\, \varsigma\, \nabla V_e(\boldsymbol{x}_a), \tag{6.7}$$

from central differences on the grid, interpolated to the atom (outside the grid: a point charge at the cloud's centroid). The forces are added to the GROMACS forces every step on every rank, and their work is booked (item 1). Between solves the ions move through the fixed electron potential.

6.7 Escape or trapping

An electron of kinetic energy ε released at atom k escapes if $\varepsilon > D_k$ and is trapped otherwise, with the depth (6.6) of the same smoothed ion potential and the same cloud the solve uses. The atom's own new charge is left out: the electron leaves before the atom's remaining electrons rearrange. An escaped electron adds one to the net charge Q; a trapped one adds one to N and is booked as item 2. Each electron is tested once, at birth: an escaped one cannot be recaptured, and a trapped one leaves only by evaporation.

6.8 Evaporation

A well of finite depth cannot hold the tail of a Maxwell distribution: an electron whose kinetic energy exceeds its local depth $D(\boldsymbol{x}) = -V(\boldsymbol{x})$ is unbound and leaves, and collisions keep refilling the tail. This evaporation (`mcmd-evaporation = 1`, the default) matters late in the explosion, when the well is shallow and the cloud still hot. For a Maxwell distribution at T, the fraction of electrons above D and the fraction of their kinetic energy above D are regularized incomplete gamma functions of $x = D/T$ [21]:

$$Q(\tfrac{3}{2}, x) = \operatorname{erfc}\sqrt{x} + 2\sqrt{x/\pi}\, e^{-x}, \qquad Q(\tfrac{5}{2}, x) = Q(\tfrac{3}{2}, x) + \tfrac{4}{3\sqrt{\pi}} x^{3/2} e^{-x}. \tag{6.8}$$

The energy tail is the larger (Fig. 6.2), so evaporation cools the cloud. Summed over the cells of the last solve (binned by depth into 4000 bins once per solve), $\delta N^{\mathrm{tail}} = \sum_i n_i Q(\frac{3}{2}, x_i)$ and $\delta E^{\mathrm{tail}} = \frac{3}{2} T \sum_i n_i Q(\frac{5}{2}, x_i)$. The tail refills at the electron–electron collision rate [17]

$$\nu_{ee} = 2.91 \times 10^{-6}\, n_e[\mathrm{cm}^{-3}]\, \ln\Lambda\, T^{-3/2}\ \mathrm{s}^{-1}, \qquad \ln\Lambda = \max\left(2,\ 24 - \ln(\sqrt{n_e}/T)\right), \tag{6.9}$$

at the mean density $n_e = N/(\frac{4}{3}\pi R^3)$, and every MD step a fraction $\phi = \min(1, \nu_{ee}\Delta t)$ of it leaves: N and K fall by $\phi\varsigma\, \delta N^{\mathrm{tail}}$ and $\phi\varsigma\, \delta E^{\mathrm{tail}}$, the net charge Q rises, and the leavers' total energy is booked (item 5). How many electrons evaporate depends on the box (Appendix A); the total that escapes, at birth and by evaporation, does not.

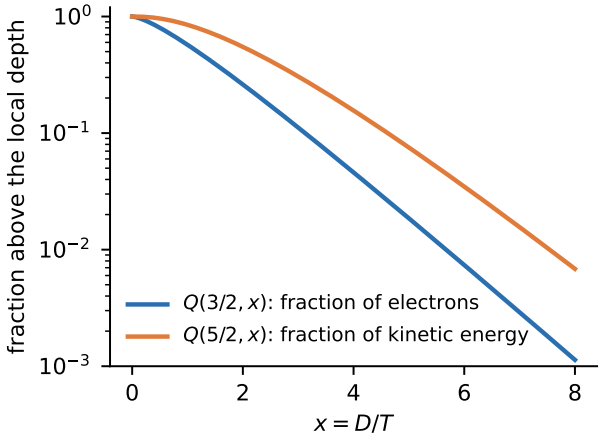


Figure 6.2: The Maxwell tails above a depth D: the fraction of electrons $Q(\frac{3}{2}, D/T)$ and of their kinetic energy $Q(\frac{5}{2}, D/T)$.

6.9 Assumptions

- One temperature, instantaneous equilibrium: hot spots are averaged out at the next solve.
- Classical mean field: no degeneracy, exchange or correlation below the scale s_e; s_e has a physical size but is not derived from first principles.
- Electrons frozen between solves (0.2 fs), adequate while the ions move much less than a grid cell in that time.
- Escape at birth only; evaporation at the mean electron–electron collision rate, an estimate rather than a kinetic calculation.
- Far from the sample the cloud is a point charge.

Chapter 7

Electron–ion collisions and field ionization

7.1 Electron-impact ionization

With the electron cloud on, its electrons ionize the atoms they reach (`mcmd-collisions = 1`, the default). Every configuration can lose one electron from each occupied shell, to any final configuration that has data. The rate of a channel $i \to f$ at atom k is

$$w_{i\to f} = n_e(\boldsymbol{x}_k)\,\langle\sigma v\rangle_{i\to f}(T), \tag{7.1}$$

with the cloud's temperature T and its density at the atom, interpolated from the last solve, $n_e(\boldsymbol{x}_k) = \varsigma \sum_{\boldsymbol{o}} w_{\boldsymbol{o}}\, n_{\boldsymbol{i}_0+\boldsymbol{o}}/\Delta V$ (cloud-in-cell weights). An ion deep in the quasi-neutral core sees a high density and ionizes quickly; one in the depleted outer sheath sees almost none. The collisional channels join the atom's other transitions in the Monte Carlo step, with the rates frozen for the step, and the energy is booked as in Section 6.5 (item 3). The transition energy is always $\Delta E = T_i - T_f$ from the total energies.

7.1.1 Rate coefficients: Lotz (built in) or from files

`mcmd-coll-data` chooses where $\langle\sigma v\rangle$ comes from.

`lotz` (default): Lotz's Maxwellian formula [7] (with $b = 0$),

$$\langle\sigma v\rangle = 6.7\times10^{-7}\, a\, q\, T^{-3/2}\,\frac{E_1(P/T)}{P/T}\ \ \mathrm{cm^3\,s^{-1}}, \qquad a = 4.5\times10^{-14}\ \mathrm{cm^2\,eV^2}, \tag{7.2}$$

q the occupation of the shell, $P = \Delta E$ (eV), E_1 the exponential integral. No data file is needed.

`files` : from `collisional_ionization_table_X.txt` if present, a table of $\langle\sigma v\rangle$ on a temperature grid that any atomic code can supply; else from the Sampson fits in `collisional_parameters_X.txt` (CRETIN models):

$$\langle\sigma v\rangle = 1.0899\times10^{-6}\,\frac{s_0\, q(b)}{\Delta E\sqrt{T}}\,e^{-b}, \quad q(b) = c_0+c_1\sqrt{\pi b}\,\mathrm{erfcx}\sqrt{b}+(c_2 b+c_3)\,e^{b}E_1(b), \quad b = \frac{\Delta E}{T}. \tag{7.3}$$

Tables are interpolated linearly in $\ln T$ in the smooth quantity $\ln\langle\sigma v\rangle + \Delta E/T$ and held at the grid edges outside them.

For H, C, N, O and S Lotz gives the same transitions as the CRETIN fits, with coefficients 13–25 % higher between 10 and 300 eV. On lysozyme at 600 eV this changes the ion kinetic energy by −2.5 % and nothing else beyond the seed noise. A `.tpr` older than format 79 runs as `files`.

7.2 Three-body recombination

The inverse of collisional ionization, two cloud electrons colliding at an ion with one of them captured, follows by detailed balance with the statistical weights g:

$$w_{f\to i} = n_e^2 \,\langle\sigma v\rangle_{i\to f}\, \frac{g_i}{g_f}\, 1.656\times 10^{-22}\, T^{-3/2}\ \ \mathrm{s}^{-1}. \tag{7.4}$$

It is off by default (`mcmd-coll-recombination = 0`) and **not fully validated**: in the classical continuum picture it recaptures far too many electrons in the dense, cool cloud late in the explosion. It grows as n_e^2/T at small T and diverges as $T\to 0$; it is therefore evaluated at $\max(T, T_{\min})$, $T_{\min}$ = `mcmd-te-min` (1 eV), which caps it there without switching it off (Section 6.4). `mcmd-coll-h-recombination` additionally allows recombination onto hydrogen (off: a proton stays a proton).

7.3 Collisional excitation

With `mcmd-coll-excitation = 1` the cloud also excites atoms between configurations with the same number of electrons, and de-excites them by detailed balance ($\langle\sigma v\rangle_{\mathrm{down}} = \langle\sigma v\rangle_{\mathrm{up}}\, g_{\mathrm{lower}}/g_{\mathrm{upper}}$). The rates come from `collisional_excitation_table_X.txt` or the CRETIN `colex2` fits in `collisional_excitation_X.txt` (Chapter 8). It is off by default and **not fully validated**: it produces too much early collisional ionization out of excited configurations.

7.4 Field ionization

In a strongly charged sample the collective field can lower an atom's barrier below its outermost binding energy. With `mcmd-field-ionization = 1` an atom is stripped while the field at it exceeds the Bethe over-the-barrier threshold,

$$F \geq \frac{I_p^2}{4k_{\mathrm{C}}Z}, \tag{7.5}$$

I_p the binding energy of its outermost electron and Z its charge after the step. $F = |\nabla P_L|$ is the field of the collective potential $P_L = k_{\mathrm{C}} G_L * (q_{\mathrm{ion}} - n_e)$, with every charge spread over L (`mcmd-field-ionization-smoothing`, 0.5 nm; at least the grid's s) and the atom's own charge left out, so neighbours closer than $\sim L$ are smeared: their barrier lowering is the charge-transfer channel's. It is computed on the grid after every solve, by direct summation where there is no grid, during the pulse only (from $6\sigma_t$ before to $4\sigma_t$ after the peak, stopping once no atom has been within a factor 2 of its threshold for 5 updates). Every field ionization is logged in `pbfield_events.txt`.

Field ionization does not change the explosion: on ubiquitin at 600 eV it strips ∼900–1000 atoms per run early in the pulse, and about as many fewer are ionized by electron impact and charge transfer; the charge, the escape and the ion directions are unchanged. It is off by default.

Chapter 8

Atomic data

By default the module computes the atomic physics of every element at start-up from a screened-hydrogenic shell model, for the photon energy of the pulse (`mcmd-atomic-data = hydrogenic`, Section 8.1); no input file is needed. With `mcmd-atomic-data = files` it reads the same quantities from plain-text tables in `./Atomic_data/` of the run directory instead. This chapter specifies the tables; `ATOMIC_DATA.md` in the repository is the same specification in short form, and `tools/check_atomic_data.py` checks a directory against it.

8.1 The built-in data

The built-in data are a screened-hydrogenic model of shells only [8], with the screening constants of Marchand, Caille and Lee [9] and the level energies scaled to the NIST ionization energies [10]:

- states: every `K L M` of every charge state ($K \leq 2$, $L \leq 8$, $M \leq 18$), so elements up to $Z = 18$ (H to Ar except He); Ca, Fe, Ni and I need `mcmd-atomic-data = files`;
- energies: the screened-hydrogenic binding energy of each configuration (with the leading relativistic terms), the excitation energy scaled by the ratio of the NIST to the hydrogenic ionization energy of the charge state its lowest inner-shell hole belongs to;
- photoionization out of each occupied shell: Kramers' cross section with a constant Gaunt factor, $\sigma \propto q^4 n^{-5} E^{-3}$ per electron (q the screened charge of the shell);
- fluorescence: the Einstein A of each radiative transition into a lower shell, from hydrogenic oscillator strengths scaled with the screened charges;
- Auger: the hydrogenic formula of Scott and Hansen [8] (Gaunt factor 0.12), summed per final state;
- statistical weights $g = \binom{2}{K}\binom{8}{L}\binom{18}{M}$.

It takes well under a second. The model is the one of the CRETIN screened-hydrogenic models the published results were computed with; it reproduces their rates to ∼10 % and, on lysozyme at 1000 eV, their explosion: charge to 0.7 %, escape to 1.3 %, directions within the seed noise, ion kinetic energy 3.5 % lower (Section 8.5). There are no collisional fit data: electron-impact ionization comes from Lotz's formula (`mcmd-coll-data = lotz`), and collisional excitation (`mcmd-coll-excitation`) and `mcmd-coll-data = files` need `mcmd-atomic-data = files`; `grompp` stops otherwise. The run writes the data it used to `simulation_output/Atomic_data/`, in the format below; copied to `./Atomic_data` and run with `mcmd-atomic-data = files` they give the identical run.

8.2 Which files

A file is `<table>_<SYMBOL>.txt`, with the symbol in upper case as in Table 3.1 (`energy_levels_C.txt`, `rate_transitions_CL.txt`). Only the elements present in the system are needed.

file	needed when (`mcmd-atomic-data = files`)	what for
`energy_levels_X.txt`	always	outermost binding energy per state
`rate_transitions_X.txt`	always	photoionization, Auger, fluorescence
`total_energies_X.txt`	charge transfer or collisions on (both default)	electron energies, charge-transfer energies, collisional transition energies
`statistical_weight_X.txt`	collisions on	detailed balance
`collisional_ionization_table_X.txt` or `collisional_parameters_X.txt`	only with `mcmd-coll-data = files`	electron-impact ionization
`collisional_excitation_table_X.txt` or `collisional_excitation_X.txt`	only with `mcmd-coll-excitation = 1`	collisional (de-)excitation

Only `rate_transitions_X.txt` depends on the photon energy (the photoionization cross sections): regenerate it for every `mcmd-pulse-photon-energy`.

General rules. One record per line, whitespace separated; `;` separates the initial state from the final states in the transition tables; no comments or header (except the tables' temperature line). Lines must be shorter than 1024 characters. Energies in eV, rates in s^{-1}, cross sections in cm^2. Where a state appears twice, the first line counts.

8.3 The tables

`energy_levels_X.txt`: `K L M E`, the binding energy of the state's *outermost* electron. Autoionizing states (excitation above the ionization potential of their charge state) get `1e14`. Used as the charge-transfer donor energy and the field-ionization threshold.

`total_energies_X.txt`: `K L M T`, the energy to remove all of the state's electrons, $T = \sum$(ionization potentials up to this charge state) − excitation energy; defined for autoionizing states too, and `0 0 0 0.0`. It must reproduce `energy_levels` for every bound state, $E(c) = T(c) - T(c$ minus its outermost electron), and cover every reachable state. It gives the kinetic energy of emitted electrons (photoelectron $E_{\mathrm{ph}} - (T_{\mathrm{from}} - T_{\mathrm{to}})$, Auger $T_{\mathrm{to}} - T_{\mathrm{from}}$, both must be positive), the binding energy of any shell for charge transfer, and collisional transition energies.

`rate_transitions_X.txt`: one line per initial state, then one `;`-terminated token per final state with five fields:

```
K L M ;K' L' M' rate type ;K'' L'' M'' rate type ;...
1 1 0 ;1 0 0 7.416614e-21 2 ;0 1 0 0.0 2 ;2 0 0 7.291753e+11 1 ;      (oxygen, 600 eV)
```

Type 0: Auger–Meitner decay, rate in s^{-1}, one electron fewer. Type 1: fluorescence, rate in s^{-1}, same electron count. Type 2: photoionization, cross section in cm^2, one electron fewer. The type column is mandatory: many Auger and photoionization transitions have the same occupation change. Every state reachable through any transition needs a line of its own, also

without transitions (`K L M ;`); only `0 0 0` may be absent. A zero rate is inactive; two entries to the same final state are two channels.

`statistical_weight_X.txt`: `K L M g`, the degeneracy $g = \binom{2}{K}\binom{8}{L}\binom{18}{M}$.

Collisional data (only for `mcmd-coll-data = files` and collisional excitation):

- `collisional_ionization_table_X.txt`, `collisional_excitation_table_X.txt`: a first line `T_e t1 ... tn` (eV, increasing, full precision), then `K L M K' L' M' k1 ... kn`: the full Maxwellian rate coefficient $\langle\sigma v\rangle$ ($\mathrm{cm^3\,s^{-1}}$, Boltzmann factor included) at each temperature. Any exponent is allowed (`3.1e-2650`); a `0` means too small to write and is extrapolated. The tables are generator independent: FLYCHK, FAC, BEB or Lotz can supply them.
- `collisional_parameters_X.txt` (`K L M ; K' L' M' c0 c1 c2 c3 s0 ; ...`) and `collisional_excitation_X.txt` (`... c0 c1 c2 c3 a0 ...`): the CRETIN Sampson and `colex2` fits, used where no table is given (formulas in Chapter 7).

8.4 Generating the data

Two generators write all the tables above, for `mcmd-atomic-data = files`.

From XATOM (`tools/make_atomic_data_xatom.py OUTDIR ENERGY [--elements H,C,N,O,S] [--validate]`). XATOM [3, 4] is an open-source (GPLv3) Hartree–Fock–Slater atomic code; its output is not covered by its license, so the generated data can be distributed freely. The script runs `xatom` once per configuration (∼20 s for H, C, N, O, S on 12 cores):

- states: every `K L M` up to the ground-state occupation — photoionization, Auger, fluorescence and charge transfer never leave that set;
- each state is computed as its aufbau subshell configuration (2s before 2p, 3s before 3p), and XATOM's subshell final states are summed back onto `K L M`;
- energies from the orbital eigenvalues, $T(c) = T(c \text{ minus its outermost electron}) - \varepsilon_{\text{outermost}}(c)$: XATOM's total energies are tens of eV off ($\mathrm{N^{5+}}$ $1s^2$: 1267 eV against 1218), its eigenvalues are what it prices transitions with;
- core and valence photoionization (`P`, `V`) become type 2, `A` type 0, `F` type 1; photoexcitation is dropped; channels these energies close and intra-shell ones are dropped and counted;
- elements up to $Z = 18$ (H to Ar except He); Ca, Fe, Ni are refused (their 4s electrons are counted in M).

From CRETIN models (`tools/make_atomic_data_cretin.py MODELS OUTDIR ENERGY [--tables] [--validate]`). Reads CRETIN `.z01` atomic models and writes all tables including the Sampson and `colex2` fits; `--tables` also writes them as temperature tables (`tools/make_collisional_tables.py`). CRETIN is not open source.

Checking. `tools/check_atomic_data.py Atomic_data --photon-energy E` reports everything the reader would stop on or misread as an error and everything legal but suspicious (rates out of range, closed channels, unreachable states) as a warning; it exits 1 on errors.

8.5 Accuracy of the data

Against Scofield's subshell photoionization cross sections [11] and Krause's K-shell widths and fluorescence yields [12] (via xraylib):

ground state, 600 eV, kb	Scofield	XATOM	CRETIN
C K / L	167.8 / 7.86	167.8 / 7.86	179.5 / 1.61
N K / L	275.9 / 14.4	276.0 / 14.4	317.6 / 4.13
O K / L	411.3 / 24.5	411.5 / 24.5	507.9 / 9.05
S L / M	445.9 / 26.6	444.7 / 26.5	419.2 / 2.19

XATOM agrees with Scofield to 0.3 % and with Krause for the S K hole (width 0.56 against 0.52 eV, fluorescence yield 0.089 against 0.080). The CRETIN models' K-shell cross sections are 7–23 % high, their valence ones 3–12× low, and their S K-hole Auger rate 3.4× low. XATOM's weak point is the outermost binding energy of neutral C and N (∼20 % low, open 2p shells). On lysozyme at 600 eV, XATOM data give a 15 % higher ion kinetic energy than CRETIN data, 1 % more charge, and the same ion directions; at 1000 eV (the example) 39 % higher, with half the Auger decays and twice the valence photoionizations.

The built-in data are of the CRETIN kind: core photoionization within 10 % of the CRETIN models, valence within 5 %, Auger and fluorescence per configuration within ∼10 %, so with the same deviations from the reference data; their K-hole Auger rates are 1.5–2.5× slower than CRETIN's, and the constant Gaunt factor makes the photoionization rise above CRETIN's towards 10 keV (C, N, O core 1.3× at 12 keV).

Chapter 9

Output

All standard GROMACS output (`.trr`, `.edr`, `.log`, `.gro`) is written as usual. The ionization module writes into `simulation_output/` in the run directory; a 30 fs lysozyme run writes about 0.6 MB there. Text files are whitespace-separated columns (`numpy.loadtxt` reads them; `#` lines are headers); times are in ps. The time-resolved files are written every `nstenergy` steps, the electron model's at every solve; all are buffered and flushed every 1000 steps.

9.1 Always written

`charges.txt` the final charge of every atom: atom number (1-based), charge.

`procces_statistics.txt` event counts at the end: Auger decays, fluorescence, photoionizations, charge transfers. (Electron-impact ionizations and recombinations are in `collisional_log.txt` and in the end-of-run summary on stderr.)

`masses.bin` the atom masses (float32), from which `tools/mcmd_output.py` gets the elements.

`mcionize_state.bin` the restart state, written at every checkpoint (Section 2.6); the previous one is kept as `mcionize_state_prev.bin`.

`timing_mcionize.txt` wall time per part of the module (Monte Carlo, charge transfer, PB solves, …; with the grid long range also the share of the long-range ion field).

`autostop_dirs.txt` with `mcmd-autostop = 2`: one line per direction check, t in fs, the median heavy-ion turn since the last check (degrees), the turn predicted still to come (-1 before there is a prediction), the number of heavy ions counted (Section 2.5). Times here are in fs.

9.2 Time-resolved output (`mcmd-detailed-output = 1`)

`mean_charge_vs_time.txt` t, mean charge per atom.

`pulse_profile.txt` t, pulse intensity $I(t)$ in photons $\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$.

`electron_data.txt` t, mean density of the trapped electrons (cm^{-3}), their temperature (eV), and the expansion $R_g/R_g(0)$. The temperature is `nan` before the first electron is trapped.

`charges_over_time.bin` every atom's charge (float32), one frame every `mcmd-charge-output-stride` steps (default 500, i.e. 1 fs at 2 as).

`transitions.bin` every electronic event: a 16-byte header (`MCMDTRN1`, version, record size), then 20-byte records: time (fs, float32), atom (0-based), the charge-transfer donor (else -1), the type (0 Auger, 1 fluorescence, 2 photoionization, 3 collisional ionization, 4 recombination, 5 excitation, 6 de-excitation, 7 charge transfer), and the `K L M` occupations

before and after. It is front-loaded: photoionization follows the pulse and the Auger cascades follow within a few fs.

9.3 The electron model (`mcmd-electrons = pbgrid`)

`pb_log.txt` one line per solve: t, N trapped, Q escaped, T, grid spacing, ions outside the grid, Newton and CG iterations, residual, wall time (ms), mean and maximum depth, the field's work and the change of U since the last solve, the kinetic energy, any clamped energy, the core neutralization (electrons per ion charge within $R/2$), convolutions, rebuilds, temperature iterations.

`screening_log.txt` every `nstenergy` steps: escaped and trapped electrons, trapped fraction, mean energy of the trapped electrons, R, Debye length and number, evaporated electrons, T (the actual one, may be below `mcmd-te-min` or negative), the work on the ions, the smoothing width, and the temperature of the solve and recombination, $\max(T, T_{\min})$.

`collisional_log.txt` every `nstenergy` steps: T, mean n_e, N, the cloud's energy, cumulative electron-impact ionizations and recombinations, the energy spent on ionization, the energy brought in by trapped electrons, the adiabatic and evaporated energy, R, Q. It is written before each step's Monte Carlo, so the last line misses the last step's events; the end-of-run summary has the final counts.

`energy_ledger.txt` cumulative energy exchanges: the ion–ion and ion–electron energy jumps of charge changes, the field's work on the ions, the electrons' kinetic energy, N.

`pbg_box.txt` the grid box at each solve: R, half-width in units of R, spacing, the estimated fraction of electrons outside, whether the grid was rebuilt, cells per side.

`pbg_lr.txt` with long range: the refreshes of the long-range ion field.

`pbfield_events.txt` with field ionization: every event (t, atom, mass, charge after, distance from the centre, field, I_p, depth).

9.4 The output viewer

`tools/mcmd_output.py` reads all of the above:

```
mcmd_output.py transitions [dir] [--type photo,auger] [--atom 12] [--tmin 20 --tmax 40]
mcmd_output.py summary [dir]           # events per type and element, and per 5 fs
mcmd_output.py charges [dir]           # mean charge per element over time
mcmd_output.py plot [dir]              # overview.png of every metric the run wrote
mcmd_output.py plot [dir] --gif        # + explosion.gif: three views along the initial
    gyration axes
mcmd_output.py plot [dir] --extract    # + extracted/: transitions.txt, charges_over_time.txt
```

`dir` is the run directory or its `simulation_output` (default: the current directory). From Python, `import mcmd_output` gives `read_transitions`, `read_charges`, `elements`, `masses` and `read_trr` (pure Python) as numpy arrays.

Chapter 10

Reference: mdp options

All options of the model are `mcmd-*` entries in the `.mdp` file. They take effect only with `mdrun -ionize`; without it the run is stock GROMACS 4.5.4 whatever the `.mdp` says. `grompp` rejects unknown `mcmd-*` keys and the removed ones, with the replacement. The defaults are the validated model; settings marked "not fully validated" are listed by the run summary as warnings (Section 2.3). Times of the pulse and of the electron model are in **fs**, unlike `dt` and `tinit`.

10.1 The pulse

option	default	unit	meaning
`mcmd-pulse-peak-time`	0	fs	centre of the Gaussian pulse
`mcmd-pulse-fwhm`	0	fs	full width at half maximum
`mcmd-pulse-photons`	0		photons in the pulse; 0: no ionization (warning)
`mcmd-pulse-focal-diameter`	0	nm	diameter of the focal spot, uniform over the sample
`mcmd-pulse-photon-energy`	0	eV	photon energy; the built-in data are computed for it, files must be generated for it
`mcmd-pulse-bandwidth`	0.003		relative FWHM of the photon energy: jitters the photoelectron energies

10.2 Atomic data

option	default	unit	meaning
`mcmd-atomic-data`	hydrogenic		`hydrogenic`: built in, screened-hydrogenic shells, H to Ar except He, no input file (Section 8.1); `files`: the tables in `./Atomic_data`

10.3 Charge transfer

option	default	unit	meaning
`mcmd-charge-transfer`	1		over-the-barrier charge transfer (Chapter 4)
`mcmd-ct-h-donor-only`	1		only neutral hydrogen donates (the published model); 0 is not fully validated
`mcmd-allow-h-ct`	0		allow H–H transfer (resonant); not fully validated
`mcmd-charge-transfer-downhill`	0		require the transfer to be downhill, Coulomb release included
`mcmd-charge-transfer-idle`	2000	steps	steps without a transfer before the pass goes idle; negative: never
`mcmd-charge-transfer-recheck`	100	steps	interval of the idle pass

10.4 The electron cloud

option	default	unit	meaning
`mcmd-electrons`	pbgrid		`pbgrid`: the Poisson–Boltzmann cloud (Chapter 6); `none`: every electron leaves
`mcmd-pb-smoothing`	0.25	nm	the electron size s_e; the ion kinetic energy depends on it (warning if changed)
`mcmd-pb-interval`	0.2	fs	time between PB solves (and long-range refreshes); converged at 0.2
`mcmd-pb-newton-tol`	0.01		Newton stop: relative residual; converged at 0.01
`mcmd-pb-cells-per-smoothing`	2.5		target spacing s_e/this at the start; converged at 2.5
`mcmd-pb-box-cut`	0.02		the box grows when more than this fraction of the electrons would be outside; 0: fixed box of `mcmd-pb-box-max`
`mcmd-pb-box-max`	3	R	largest box half-width
`mcmd-pb-grid-limit`	384	cells	most cells per side; a warning in `md.log` says when a sample needs more
`mcmd-pb-long-range`	auto		`auto`: above 10 000 atoms a 2 nm cutoff plus the long-range ion force from the grid, else all pairs (`grompp` sets the cut-offs); `1/0` keep the mdp's cut-offs
`mcmd-evaporation`	1		evaporation of the cloud; 0 is not fully validated

option	default	unit	meaning
`mcmd-te-min`	1	eV	temperature floor of the PB solve, the evaporation and three-body recombination; the actual T is tracked below it, ionization uses it; 0: the release model (Section 6.4)

10.5 Collisions and field ionization

option	default	unit	meaning
`mcmd-collisions`	1		electron-impact ionization in the cloud (Chapter 7)
`mcmd-coll-data`	lotz		`lotz`: built-in formula; `files`: tables or fits from `Atomic_data`
`mcmd-coll-recombination`	0		three-body recombination; not fully validated
`mcmd-coll-h-recombination`	0		recombination also onto hydrogen
`mcmd-coll-excitation`	0		collisional excitation and de-excitation; not fully validated
`mcmd-field-ionization`	0		over-the-barrier ionization in the cloud's field
`mcmd-field-ionization-smoothing`	0.5	nm	L of the collective field

`mcmd-collisions`, `mcmd-field-ionization` and `mcmd-pb-long-range = 1` need the cloud, and `mcmd-coll-data = files` and `mcmd-coll-excitation = 1` need `mcmd-atomic-data = files`; `grompp` stops otherwise.

10.6 Run control and output

option	default	unit	meaning
`mcmd-seed`	-1		random seed; -1: new each run, printed in the summary
`mcmd-detailed-output`	1		the time-resolved output (Chapter 9)
`mcmd-charge-output-stride`	500	steps	frames of `charges_over_time.bin`
`mcmd-autostop`	0		end the run early (`nsteps` stays the limit): 0 off; 1 once $E_{\mathrm{kin}}/E_{\mathrm{tot}}$ exceeds `mcmd-autostop-energy-ratio`; 2 once the ion directions have settled (Section 2.5)
`mcmd-autostop-energy-ratio`	0.99		the ratio of `mcmd-autostop = 1` (was `mcmd-autostop-threshold`)
`mcmd-autostop-angle`	2	deg	`mcmd-autostop = 2`: stop once the median heavy-ion turn still to come is predicted below this
`mcmd-autostop-interval`	5	fs	time between the direction checks
`mcmd-autostop-min-time`	60	fs	no direction stop earlier than this after the pulse peak

option	default	unit	meaning
`mcmd-schedule`	(empty)	fs	a larger MD step and PB interval later in the run, as `t:dt:pb` entries (Section 2.5); empty: one step size throughout
`mcmd-initial-charges`	0		read starting charges from `charges.txt` (atom, charge)

10.7 GROMACS settings for an explosion run

- `dt = 0.000002` (2 as): the model is converged at ≤2 as (warning above); after the pulse `mcmd-schedule` can raise it (Section 2.5);
- `morse = yes`: bonds that can break (Chapter 5);
- `pbc = no`, `coulombtype = Cut-off`, no temperature or pressure coupling;
- cut-offs: left to `mcmd-pb-long-range = auto`;
- `nstenergy`: the interval of the module's time-resolved text output.

10.8 Renamed and removed options

Earlier versions used GROMACS's generic `userint`/`userreal` slots; `grompp` rejects them and names the replacement:

old	new
`userint1`	removed: `-ionize` alone enables the modified force field
`userint2`	`mcmd-charge-transfer`
`userint3`	`mcmd-autostop`
`userint4`	removed: use `mdrun -cpi`
`userint5`	`mcmd-detailed-output`
`userint6`	removed (never implemented): electron-impact ionization is `mcmd-collisions`
`userint7`, `userint8`	`mcmd-charge-transfer-idle`, `-recheck`
`userint9`	`mcmd-initial-charges`
`userreal1`	`mcmd-pulse-peak-time`, now in fs (multiply by 1000)
`userreal2`	`mcmd-pulse-photons`
`userreal3`	`mcmd-pulse-fwhm`, now the FWHM in fs (multiply by 2354.82)
`userreal4`, `userreal5`	`mcmd-pulse-focal-diameter`, `-photon-energy`
`userreal6`	`mcmd-autostop-energy-ratio`

`mcmd-autostop-threshold` is now `mcmd-autostop-energy-ratio` (same meaning); `grompp` names the new key. The development versions of the electron model were controlled by `GMX_MCMD_*` environment variables. Those that became options (`GMX_MCMD_SCREEN`, `_SEED`, `_PB_NSTEP`, `_PBG_SE`, …) and those that were removed stop `mdrun` with a message naming the replacement, so an old launcher cannot run a different model unnoticed.

Chapter 11

Reference: diagnostic environment variables

The model itself is set entirely in the `.mdp`. A few environment variables remain, for diagnostics only: they add output or checks, and the physics is the same with or without them (except for the thread count of the FFTs, which can change the last bits of the solve). Set them to 1 to switch them on.

variable	effect
`GMX_MCMD_TIMING_DETAIL`	`pbg_timing.txt`: per solve, the Newton and CG iterations, convolutions, and the time in FFTs, convolutions, CG and Newton; the residuals. Small overhead.
`GMX_MCMD_PB_EDIAG`	`pb_ediag.txt`: per solve, the split of the electrons' energy (kinetic, electron–ion, electron–electron, the exchanges since the last solve), for checking the energy bookkeeping.
`GMX_MCMD_PBG_DENSITY`	`pbg_density.bin`: the grid density every tenth solve (header t, N, T, Δx, origin, n_g, then n_g^3 floats in nm^{-3}). Large: hundreds of MB to GB per run.
`GMX_MCMD_COLL_EVLOG`	`coll_event_log.txt`: every collisional event with the local density and temperature. Large.
`GMX_MCMD_PBG_OUTLAT_CHECK`	compares the sub-lattice potential of the ions outside the box with the exact sum (`pbg_outlat.txt`) and logs the ions and time per stride level (`pbg_outpot.txt`). Slow.
`GMX_MCMD_PBG_LR_CHECK`	with long range: compares the grid's long-range ion field with the exact all-pairs field at every refresh (`pbg_lr.txt`). $O(N^2)$ on the master.
`GMX_MCMD_PD_CHECK`	under particle decomposition: every 50 steps, checks each rank's coordinates against the master's copy (the master needs all coordinates for the solve).
`GMX_MCMD_PBG_FFT_THREADS`	threads for the FFTs of the solve; default the number of ranks (Appendix B).

Every `GMX_MCMD_*` variable of the development versions that is not in this table is refused: `mdrun` stops and names the mdp option that replaced it, or says that it was removed.

Appendix A

Numerics of the electron model

This appendix describes how the free energy of Chapter 6 is discretized and minimized, and the devices that keep the solve affordable for 10^3–10^5 atoms: an adaptive grid, an adaptive box, sub-lattices for distant ions, and the long-range ion force from the grid. The code is in `src/kernel/mcionize.c`.

A.1 The grid

The grid is a cube of n_g^3 nodes with spacing Δx and half-width h, centred on the heavy-atom centre of mass $\boldsymbol{c}$: $\boldsymbol{x}_{ijk} = \boldsymbol{c} - h(1,1,1) + \Delta x\,(i,j,k)$, $\Delta x = 2h/(n_g - 1)$. The variables are the electron counts n_i at the nodes, $\sum_i n_i = N$, the density $n_i/\Delta V$. The sample radius $R = (a_1 a_2 a_3)^{1/3}$ is that of the heavy-atom, mass-weighted gyration ellipsoid ($a_\alpha = \sqrt{5\lambda_\alpha}$), the radius of a uniform sphere; hydrogens are left out because the fast protons leave the cloud behind.

Each charged atom inside the grid is deposited onto its 8 surrounding nodes with cloud-in-cell (trilinear) weights [22, 23]; the same weights interpolate grid quantities back to an atom. With $K_{ij} = k_C G_s(|\boldsymbol{x}_i - \boldsymbol{x}_j|)$ and Φ_i the ions' potential at node i, the discrete free energy, its gradient and Hessian are

$$F(\boldsymbol{n}) = \sum_i n_i \Big[-\Phi_i + \tfrac{1}{2}(\mathsf{K}\boldsymbol{n})_i + T\Big(\ln\frac{n_i}{\Delta V} - 1\Big)\Big], \tag{A.1}$$

$$g_i = V_i + T\ln\frac{n_i}{\Delta V}, \quad V_i = -\Phi_i + (\mathsf{K}\boldsymbol{n})_i, \qquad H_{ij} = K_{ij} + \frac{T}{n_i}\delta_{ij}. \tag{A.2}$$

At the minimum on $\sum n_i = N$ all g_i are equal to μ.

A.2 Open-boundary convolution by FFT

Every product $\mathsf{K}\boldsymbol{a}$ is a discrete convolution, and the dominant cost. A periodic FFT of size n_g^3 would add the images of neighbouring boxes; the *aperiodic* convolution is computed exactly by Hockney's method [24, 25]: embed $\boldsymbol{a}$ in a zero-padded grid of $(2n_g)^3$, tabulate the kernel on the same doubled grid at the minimum-image separation, multiply the transforms and transform back. No two physical nodes are more than $n_g - 1$ apart, so the circular convolution on the doubled grid is the aperiodic one. The kernel transform depends only on Δx and s and is computed once per grid rebuild. FFTW [26] in single precision, real-to-complex, multithreaded; one convolution costs two FFTs of size $(2n_g)^3$.

A.3 Newton's method

The minimizer is found by Newton's method on the constraint surface [27]: the step $\boldsymbol{d}$ minimizes $\boldsymbol{g}^\mathsf{T}\boldsymbol{d} + \frac{1}{2}\boldsymbol{d}^\mathsf{T}H\boldsymbol{d}$ subject to $\sum_i d_i = 0$, i.e. $H\boldsymbol{d} + \lambda\mathbf{1} = -\boldsymbol{g}$, $\mathbf{1}^\mathsf{T}\boldsymbol{d} = 0$. H is symmetric positive

definite, and the system (n_g^3 up to 10^7 unknowns) is solved by conjugate gradients restricted to the constraint, the projected CG of Gould, Hribar and Nocedal [28], preconditioned by the diagonal of H (Jacobi): $M_{ii} = K_0 + T/n_i$, $K_0 = 2k_{\mathrm{C}}/(s\sqrt{\pi})$. The projection is built into the preconditioning step, $\boldsymbol{z} = M^{-1}(\boldsymbol{r} - \ell\mathbf{1})$, $\ell = \mathbf{1}^{\mathsf{T}}M^{-1}\boldsymbol{r}/\mathbf{1}^{\mathsf{T}}M^{-1}\mathbf{1}$, so every search direction conserves the electron count. CG stops at a relative projected residual of 10^{-4} (squared 10^{-8}), or after 500 iterations.

The step is applied multiplicatively, $n_i \leftarrow n_i \exp(\mathrm{clip}_{[-3,3]}\, \alpha d_i/n_i)$, then renormalized to N: to first order the Newton step, but every n_i stays positive. The step length α starts at 1 and is halved, at most 8 times, until the free energy *or* the residual falls. (Near the target the fall of F, $\sim NTr_{\mathrm{PB}}^2$, is below the single-precision noise of F; accepting a falling residual too avoids stalling there.)

The solve has converged when g_i is the same in every occupied cell; the code measures the density-weighted spread,

$$r_{\mathrm{PB}} = \frac{1}{T}\Big[\frac{1}{N}\sum_i n_i (g_i - \bar{g})^2\Big]^{1/2}, \tag{A.3}$$

the rms relative error of an electron's Boltzmann factor, and stops at $r_{\mathrm{PB}} <$ `mcmd-pb-newton-tol` (10^{-2}), or after 60 Newton steps. The first solve starts from the electrons deposited on the positive ions; later ones from the previous solution, rescaled to the current N (a warm start). 10^{-3} of the electrons are spread uniformly, which keeps every $n_i > 0$ and lets the cloud reach cells it had left.

A.4 The temperature iteration

T and $\boldsymbol{n}$ are found together from $g(T) = E_{\mathrm{tot}} - U(\boldsymbol{n}_T) - \frac{3}{2}NT = 0$ (Section 6.4): start from the temperature of the current kinetic energy (at least `mcmd-te-min`, or 1 eV when that is 0); solve for $\boldsymbol{n}_T$; stop if $|g| \le 0.01 \cdot \frac{3}{2}NT$; otherwise move towards $T_t = (E_{\mathrm{tot}} - U)/\frac{3}{2}N$ by at most a factor 4 until the root is bracketed, then take secant steps (the geometric mean of the bracket if the secant lands within 5 % of an end); at most 12 iterations, each warm-started.

A.5 When the solve runs

A solve runs every `mcmd-pb-interval` (0.2 fs, i.e. every 100 steps at 2 as), and also whenever the number of trapped electrons has grown by more than 20 % (plus 10) since the last one, so that the cloud keeps up during the pulse. 0.2 fs gives the 0.1 fs results within the seed noise in ion energy, charge, escape and directions; from 0.5 fs the cloud stays too hot and the ion energy drops by up to 20 %.

A.6 Adaptive grid size

With the physical floor s_e the grid only needs to resolve s_e. Each rebuild chooses n_g from the target spacing

$$\Delta x_t = \frac{s_e}{\texttt{mcmd-pb-cells-per-smoothing}}\,\max\Big(1, \frac{R}{R_0}\Big), \qquad n_g = 8\Big\lceil\frac{\lceil 2h/\Delta x_t\rceil + 1}{8}\Big\rceil, \tag{A.4}$$

R_0 the radius at the first solve, with at least 24 cells per side and at most what the sample needs at the largest box (set at the first solve), capped at `mcmd-pb-grid-limit` (384). At the start $\Delta x = s_e/2.5 < s_e/2$, so $s = s_e$; as the sample expands every structure in the density stretches with it, and the target spacing grows in proportion, so the grid coarsens instead of growing. n_g is a multiple of 8 to keep the FFT sizes efficient. A sample too large for the limit gets a coarser grid late in the run, and `md.log` says so.

A.7 Rebuilds and the adaptive box

The grid is rebuilt at a solve when R has changed by more than 10 % since the last rebuild, the centre has moved by more than one cell, or the box changes. The previous density is interpolated onto the new nodes and renormalized, the kernel transform recomputed (the FFT plans if n_g changed), and the numerical change of U moved into E_{tot}.

A box of $\pm 3R$ is safe but mostly empty. The adaptive box (`mcmd-pb-box-cut`= c, 0.02) uses the smallest half-width that leaves at most a fraction c of the electrons outside, between $1.3R$ and `mcmd-pb-box-max`$\cdot R$ ($3R$); a smaller box means a finer grid at the same cost. The electrons outside the current box are estimated from a Boltzmann tail around the net charge Q at the centre, $n \simeq A\exp(k_{\mathrm{C}}Q/(T|\boldsymbol{x}-\boldsymbol{c}|))$, with A fitted to the outermost layer of cells and integrated out to the largest box, in shells of Chebyshev distance (the box is a cube). The box grows at once, to 5 % beyond the wanted size, when more than c would be outside; it shrinks only to where at most 0.1 % are outside (*safe shrink*), so that it never cuts the cloud. How many electrons evaporate depends on the box; the total that escapes does not.

A.8 Ions outside the box

Fast protons and some heavy ions leave the box early; their attraction still acts on the cloud. Ions within 16 cells of the box are deposited on a correspondingly larger FFT grid. Farther ones are added point by point, but on *sub-lattices*: a distant ion's potential is smooth over the box, so it is summed on every S-th node, with the largest $S = 2^L$ ($L \le 4$) that satisfies $S\Delta x \le d/3$ (d its distance from the box), and interpolated trilinearly to the full grid. The error is of order $(S\Delta x/d)^2$ of an already small correction.

A.9 The long-range ion force from the grid

Without a cutoff GROMACS evaluates all $O(N_{\mathrm{at}}^2)$ ion pairs. The grid already holds the smoothed potential Φ of all ions, so above 10 000 atoms (`mcmd-pb-long-range = auto`) the long-range ion–ion force is taken from it and GROMACS computes the bare Coulomb force only within a cutoff $r_c = 2\,\mathrm{nm}$. To avoid counting the near pairs twice, the smoothed near pairs are subtracted from the grid field:

$$\boldsymbol{E}_a = -\nabla\Phi(\boldsymbol{x}_a) + \sum_{b\neq a,\ r_{ab}<r_c} k_{\mathrm{C}} q_b\, G_s'(r_{ab})\,\hat{\boldsymbol{r}}_{ab} \;\approx\; \sum_{b,\ r_{ab}\ge r_c} k_{\mathrm{C}} q_b\,\frac{\hat{\boldsymbol{r}}_{ab}}{r_{ab}^2}, \qquad \boldsymbol{F}_a = q_a\boldsymbol{E}_a, \tag{A.5}$$

accurate to $\sim \mathrm{erfc}(r_c/s)$. The atom's own deposit is corrected exactly; atoms outside the grid stencil sum their far pairs directly; excluded pairs are subtracted too. The field is refreshed every `mcmd-pb-interval` of the `.mdp` (at every regular solve without a schedule) and multiplied by each atom's current charge in between. A schedule's longer PB intervals do not apply to it: the bare far-field push falls as the sample expands, and held for 2–5 fs it over-drove the explosion (11k atoms: ion kinetic energy +17 to +50 % at 200 fs). For up to ∼10 000 atoms all pairs is as fast; for the GroEL ring (55 000 atoms) long range is 1.9× faster.

A.10 Changing the step during the run

With `mcmd-schedule` (Section 2.5) the stage boundaries are steps that follow from the `.tpr` alone, so the stage of any step is known without storing anything, and a continuation (`mdrun -cpi`) finds its stage from its first step. At the first step of a stage:

- `dt` changes; leap-frog holds $\boldsymbol{v}(t - \Delta t_{\mathrm{old}}/2)$, so before the update the velocities get $\boldsymbol{a}(t)\,(\Delta t_{\mathrm{old}} - \Delta t_{\mathrm{new}})/2$, which makes the next half step $\boldsymbol{v}(t + \Delta t_{\mathrm{new}}/2)$;

- every interval counted in steps keeps its length in time (at least one step): `nstlist`, `nstcalcenergy`, `nstcomm`, `nstenergy`, `nstlog`, the trajectory intervals, the charge output stride, the charge-transfer idle and recheck intervals;
- the PB solve moves to the stage's interval; the long-range ion field keeps the `.mdp`'s (Section A.9), and the direction checks of the autostop their interval in fs;
- a line in `md.log` gives the stage, its first step, `dt`, the PB and long-range intervals in steps and `nstenergy`.

The output intervals and `nstcalcenergy` count from `tinit` in the stage's step, so frames and text output stay on the grid $t_{\mathrm{init}} + k\,\Delta t_{\mathrm{out}}$ whatever the switch times (`grompp` checks that the intervals divide); `nstlist` and `nstcomm` count from the stage's first step. Step numbers in the `.trr` are therefore no longer proportional to time; the times are. The run summary lists the stages, and the run's step count is that of the stages up to `tinit + nsteps`×`dt`. Without a schedule the build reproduces the previous one bit for bit.

A.11 One MD step

Within one MD step the module, on the master rank,

1. updates the sample geometry (centre, gyration ellipsoid, R);
2. evaporates (if $N > 0$) with the current T and the last solve's depth histogram;
3. runs charge transfer, then the Monte Carlo step for every atom (photoionization, Auger, fluorescence, collisional ionization at T and the local n_e), with the escape test and the energy booking for every released electron;
4. books the change of the cloud–ion energy from the charge changes (the ledger);
5. if a solve is due: chooses the box and rebuilds the grid if needed; computes Φ; solves for $\boldsymbol{n}_T$ and T; computes $V_e = \mathsf{K}\boldsymbol{n}$, the densities at the atoms and the depth histogram; and broadcasts V_e;

then every rank adds the cloud's force $q_a \varsigma \nabla V_e$ (and the long-range ion force) to its atoms, the work is booked, and GROMACS integrates the equations of motion.

A.12 Cost

The PB solve is 50–80 % of the wall time. In core-hours per 10 fs simulated (2 as, PB every 0.2 fs): lysozyme (1960 atoms) 0.09–0.15, 1B7U (10 430) 1.1–1.3, the GroEL ring (54 768) 10–12 with long range (8 threads; 16 bring nothing) and 18.5 with all pairs. Figure A.1 shows the size dependence.

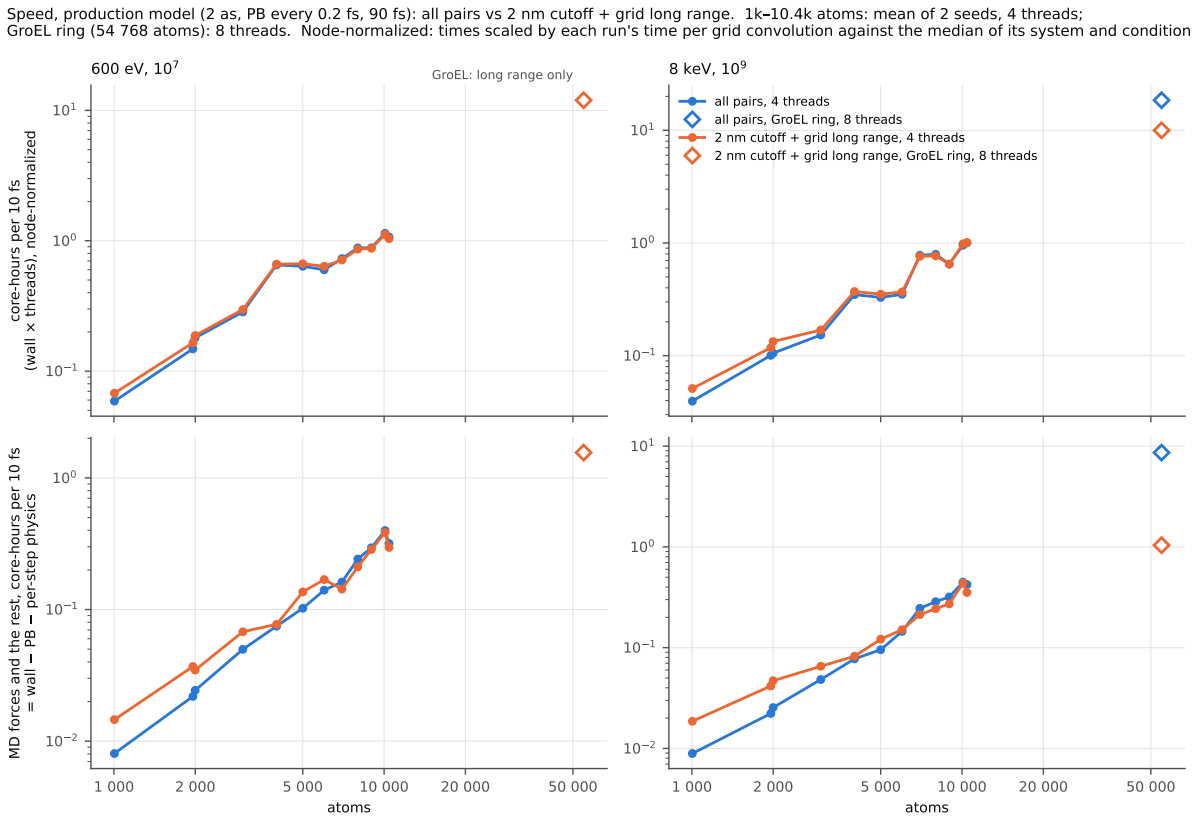


Figure A.1: Cost of the production model against system size (core-hours per 10 fs, node-normalized): all pairs against the 2 nm cutoff plus grid long range. 1k–10k atoms: 4 threads, mean of 2 seeds; GroEL: 8 threads. Bottom: the MD forces and the rest, without the PB solve.

Appendix B

Parallel runs

B.1 Particle decomposition

`mdrun -nt N -pd` runs N thread-MPI ranks with particle decomposition. Domain decomposition is refused: it renumbers atoms into rank-local indices, and the module needs global searches over all positions (charge transfer, the escape depth, the grid deposit). GPUs are not supported.

B.2 What runs where

The ionization module and the PB solve run on the master rank. To have all coordinates every step, the master gathers them over the whole ring of ranks. During the solve the other ranks sleep on a condition variable (instead of spinning in the broadcast), and their cores run the multithreaded FFTs (`GMX_MCMD_PBG_FFT_THREADS`, default the number of ranks). The master then broadcasts the electron potential (single precision) and a few scalars, and every rank adds the cloud's forces to its own atoms. The MD itself (GROMACS's forces and integration) is distributed as usual.

B.3 How many threads

The PB solve dominates (50–80 % of the wall time) and scales with the FFT threads, the rest with the ranks. 4 threads suit 1k–10k atoms; 8 suit 50k atoms; 16 brought nothing for the GroEL ring. For an ensemble it is usually more efficient to run many independent runs side by side than to parallelize one.

B.4 Reproducibility across rank counts

With a fixed `mcmd-seed` the run is reproducible on the same number of ranks and the same build. On a different number of ranks the electronic events are the same at first (the Monte Carlo runs on the master from one stream), but GROMACS sums the forces in a different order, so the positions differ in the last bits (10^{-6} nm after 2 fs on lysozyme). As in any MD this grows over a run: runs on different rank counts agree statistically, not bit for bit.

Appendix C

Validation and limitations

C.1 Numerical convergence

The production settings were checked on lysozyme (1960 atoms) and 1B7U (10 430 atoms) at 600 eV and 8 keV, with several seeds per setting:

- the MD step (converged at 2 as), the PB interval (0.2 fs), the Newton tolerance (10^{-2}), the box and the grid spacing: the charge, the escape and trapping, the radius and the ion directions are converged to within the seed-to-seed noise;
- **not converged:** the ion kinetic energy depends on the late-time smoothing — once the grid coarsens with the expanding sample, the effective s grows above s_e. At 600 eV the ion kinetic energy is up to ∼15 % above that of a run with the smoothing held at s_e, at 8 keV ∼5 %. Compare kinetic energies between runs at equal settings.

The electron size s_e: charge, escape and directions are independent of it; the ion kinetic energy at 600 eV changes by −15 % / +13–19 % for $s_e = 0.20$ / 0.30 nm.

The long-range ion force (above 10 000 atoms) leaves charge, escape and ion maps unchanged but raises the ion kinetic energy by 2–6 % against all pairs (GroEL at 8 keV: +5.3 %).

C.2 The schedule and the direction stop

`mcmd-schedule` was tested at 2 keV, 2.6×10^{10} photons into 100 nm, 10 fs FWHM peaking at 10 fs (mean heavy-atom charge 4–5), on four proteins with 4 seeds each, every scheduled run paired with an unscheduled one of the same seed (median angle of the heavy-ion velocity directions, ion kinetic energy, and 10°-smoothed ion maps, $1 - r$), at 200 fs:

	atoms	IS1b angle	IS1c angle	IS1c ΔE_{kin}	IS1c map $1 - r$	seed vs seed map $1 - r$
1Z6C	1271	0.21°	0.16°	−0.43 %	2.2×10^{-4}	2.4×10^{-2}
1G2P	2666	0.27°	0.19°	−0.66 %	4.1×10^{-4}	3.0×10^{-2}
3ICK	5748	0.52°	0.38°	−1.26 %	6.3×10^{-4}	9.1×10^{-3}
4TSN	10 991	1.25°	0.90°	−2.58 %	2.1×10^{-3}	5.6×10^{-3}
9S2H*	21 600	2.37°	1.76°	−4.2 %	4.8×10^{-3}	—

IS1c is `40:0.01:0.2 70:0.1:2 110:0.1:5`, IS1b the same with a 0.5 fs PB interval from 40 fs; *one seed, at 220 fs. The 4 seeds agree to a few hundredths of a degree, so the 28 % between the two is real. The charge does not change. The direction of one heavy ion differs between two seeds by ∼10°, and the maps by far more than the schedule moves them. The kinetic energy comes out low, and more so the larger the protein: the cloud, solved less often, holds the ions back. Most of it arises at the 70 fs change (9S2H: −0.05 % at 70 fs, −1.2 % at 75 fs). It is a delay, not lost energy: in runs to 1000 fs it fell from −0.45 % to +0.14 % (1Z6C) and from −1.23 % to

−0.02 % (3ICK); the direction error stops growing by 300 fs (1Z6C 0.19°, 3ICK 0.47°). A first change at 25 fs instead of 40 fs doubled the error at 11k atoms (3.0° against 1.5°). A further stage `300:0.5:20` changed directions by 0.01–0.06° and the kinetic energy by −0.3 % up to 1000 fs.

Wall time to 200 fs on one thread, with IS1b / IS1c: 1Z6C 2.9 / 2.6 times faster, 1G2P 3.4 / 3.1, 3ICK 3.5 / 3.3, 4TSN 2.2 / 2.2; one thread costs 27 % fewer core-hours than four (3ICK). With `mcmd-autostop = 2` as well, the schedule saves 1.3–2 times (larger proteins more); the stop comes 5 fs later with it, 25 fs for 4TSN. For the direction stop itself see Section 2.5.

C.3 Against a particle code

The model was compared with XMDYN [3], which follows every released electron as a classical particle, run without recombination, on lysozyme (8 seeds) and 1B7U:

- charge, escape, radius and ion kinetic energy agree within a few per cent at high fluence;
- at 600 eV and low fluence (10^5) the ion kinetic energy is 11 % higher and the expansion starts earlier;
- the peak electron temperature is 25–30 % lower;
- at 8 keV and 10^7 the charge is half of XMDYN's (late impact ionization in a shallow well);
- heavy-ion maps correlate $r = 0.84$–0.90 with XMDYN's, against 0.88–0.93 from seed to seed; mean directions differ by 1.5–3° beyond the seed spread.

C.4 Statistics

A single run is one realization of the Monte Carlo. A lysozyme heavy-ion map needs 3–10 seeds to correlate at $r = 0.99$ with the infinite-seed map, 1B7U one; per-atom directions scatter by 6.5–16° from seed to seed. When the start structures also differ (an ensemble of equilibrated structures, e.g. the ubiquitin variants), the map needs 150–250 runs for $r = 0.99$. The ion directions stop turning (to within 2°) 60 fs after the pulse peak (90 fs at 600 eV and low fluence).

C.5 Atomic data

The data matter: XATOM data (which match the reference photoionization cross sections) give a 15 % higher ion kinetic energy at 600 eV than the CRETIN models (whose valence cross sections are 3–12× too small), and 39 % higher at 1000 eV, with the same charge to 1–2 % and the same directions (Section 8.5). The built-in screened-hydrogenic data reproduce the CRETIN models: on lysozyme at 1000 eV the charge agrees to 0.7 %, the escape to 1.3 %, the directions within the seed noise, and the ion kinetic energy is 3.5 % lower.

C.6 Limitations

- **Elements**: up to the M shell; elements with 4s electrons count them in M, and iodine is a placeholder.
- **Configurations**: shell averaged, no subshells or levels.
- **Electrons**: one temperature, classical mean field, escape tested at birth only (Chapter 6); the ion kinetic energy depends on s_e and on the late smoothing.
- **Not fully validated**: three-body recombination, collisional excitation, H–H charge transfer, transfer from heavy donors, evaporation off.

- **Force field**: harmonic bonds are not modified (use `morse = yes`; the run summary warns).
- **Large forces**: highly charged samples produce large forces; check that the kinetic and potential energy evolve smoothly, and lower `dt` if they do not.
- Tested on Linux only.

Appendix D

Units, source files and history

D.1 Units and constants

quantity	unit	note
length	nm	GROMACS
time	ps	GROMACS; pulse and `mcmd-pb-interval` in fs
energy (module)	eV	GROMACS uses kJ/mol: 1 eV = 96.485 kJ/mol
charge	e	integer ionization charges
rates	s^{-1}	cross sections in cm^2, $\langle\sigma v\rangle$ in $cm^3\,s^{-1}$, density in cm^{-3} or nm^{-3}
$k_C = e^2/4\pi\varepsilon_0$	1.43996 eV nm	`MCMD_COULOMB_EV_NM`
pulse FWHM / σ_t	$2\sqrt{2\ln 2} = 2.35482$	

D.2 Source files

file	content
`src/kernel/mcionize.c`	the ionization module: Monte Carlo, charge transfer, the electron cloud, collisions, field ionization, output, restart state, run summary
`src/kernel/mcionize.h`	its interface to `md.c`
`src/kernel/atomic_data.c`, `.h`	the element table, the readers of `Atomic_data/`, the built-in screened-hydrogenic data
`src/kernel/md.c`	calls the module every step (`-ionize`)
`src/gmxlib/bondfree.c`	the modified bonded terms, bond breaking
`src/gmxlib/nonbonded/nonbonded.c`	the modified 1–4 Coulomb
`src/kernel/readir.c`, `grompp.c`	the `mcmd-*` options, their checks, `mcmd-pb-long-range = auto`
`src/gmxlib/tpxio.c`, `include/types/inputrec.h`	the options in the `.tpr` (format 76–83)
`tools/`	`check_atomic_data.py`, `make_atomic_data_xatom.py`, `make_atomic_data_cretin.py`, `make_collisional_tables.py`, `mcmd_output.py`
`test/regression.sh`	the regression test

D.3 Version history

tpr 73–76 the original MolDStruct options as named `mcmd-*` parameters (formerly `userint`, `userreal`).

tpr 77 the electron cloud and its options in the `.mdp` (formerly environment variables).

tpr 78 the seed and the remaining physics switches in the `.mdp`; `mcmd-pb-long-range = auto`.

tpr 79 `mcmd-coll-data`: Lotz's formula built in as the default; older `.tpr` files run as `files`.

tpr 80 `mcmd-atomic-data`: the screened-hydrogenic atomic data built in as the default; older `.tpr` files run with `./Atomic_data` (`files`).

tpr 81 `mcmd-te-min`: a temperature floor (1 eV) for the PB solve, evaporation and three-body recombination, with the actual T tracked below it and no energy created; older `.tpr` files run with 0, the model of the first release.

tpr 82 `mcmd-autostop = 2`: the run stops once the ion directions have settled (with the options `-angle`, `-interval` and `-min-time`); the energy threshold is now called `mcmd-autostop-energy-ratio`.

tpr 83 `mcmd-schedule`: a larger MD step and PB interval after the pulse; older `.tpr` files run with one step size.

The model's state at this version: binary `transitions.bin`; the run summary with warnings; built-in screened-hydrogenic atomic data by default, no input files; fluorescence rates corrected in both generators (earlier CRETIN-derived tables had fluorescence rates many orders of magnitude too small and are refused by the reader).

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